



IFM03B3

INFORMATICS

3B Mini Project A

Project: Conference Room Booking System

OVERVIEW

The Informatics 3B individual mini project serves as a replacement for the main Third Year Project. This option is only available to students that did not pass the practical component of Informatics 3B i.e. IPM (Individual Project Mark) was less than 50%.

Requirements

The goal of the mini project is to supply you with an opportunity to demonstrate to us your basic understanding of the fundamentals of control structures and problem-solving techniques.

The following criteria are applicable regardless of the project you choose:

- The project must be a Client-Server application targeting a small user base (i.e. a small business)
- The project must have a web-enabled component
- The project must have a mobile application
- The application must include data access with the following conditions:
- Create, Retrieve, Update and Delete functionality as appropriate
- Your application may use any RDBMS except MS Access
- All work progress must be committed to the provided Git repository



Objective

Develop a comprehensive system for managing conference room bookings within a single office building. This system will enable the office manager to manage different conference rooms, track bookings, and handle employee registrations. Employees will be able to search for and reserve conference rooms, while the office staff can oversee bookings, manage room availability, and generate detailed reports.

Project Description

The office manager must be able to add and manage the different conference rooms available within the building. Each room should have a description, a capacity, and a set of amenities (e.g., projector, whiteboard, video conferencing equipment). The manager should also be able to update room details and availability status. Additionally, the stock level for consumable items (e.g., markers, notepads) used in the conference rooms must be recorded, and new stock should be added when necessary.

Employees can create a booking. A single booking can include multiple conference rooms for a specific date and time. During the booking phase, the system must verify room availability and compatibility with the requested amenities. Once the employee is satisfied, they can confirm the booking, and an office clerk is then informed.

The office clerk must be able to view all bookings and the details within each booking. The system should inform them exactly which amenities are required for each booking. Once the clerk has prepared the rooms, they can finalize the booking details, and the availability status of the rooms is updated accordingly. After the conference, the clerk can reset the room's status to available.

At any time, the manager must be kept informed of the availability of conference rooms and view the usage and popularity of each room to make informed decisions.





Project Requirements:

1 Room and Amenity Management:

- The office manager can add, update, and remove conference rooms, including details like description, capacity, and amenities.
- The manager can update room availability and manage stock levels for consumable items.

2 Employee Registration and Profile Management:

- Employees can register for an account by providing their name, department, and contact information.
- Employees can view and update their profile information and booking history.

3 Room Booking:

- Employees can search for available conference rooms by date, time, capacity, and amenities.
- The system verifies room availability and allows employees to proceed with their booking.

4 Booking Management:

- The office clerk can view all bookings and the details of each booking.
- The clerk prepares the rooms with the required amenities and updates the booking status.

5 Reporting and Analytics:

- Managers can generate reports on room availability and usage.
- The system provides insights into the popularity and usage trends of each conference room to help managers make informed decisions.

